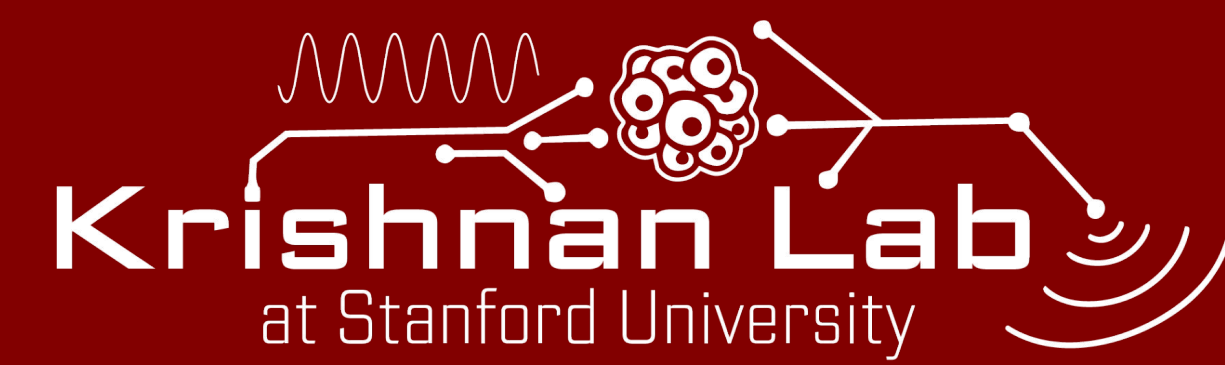




Battery-free oxygenation for long-term cell therapies in mice

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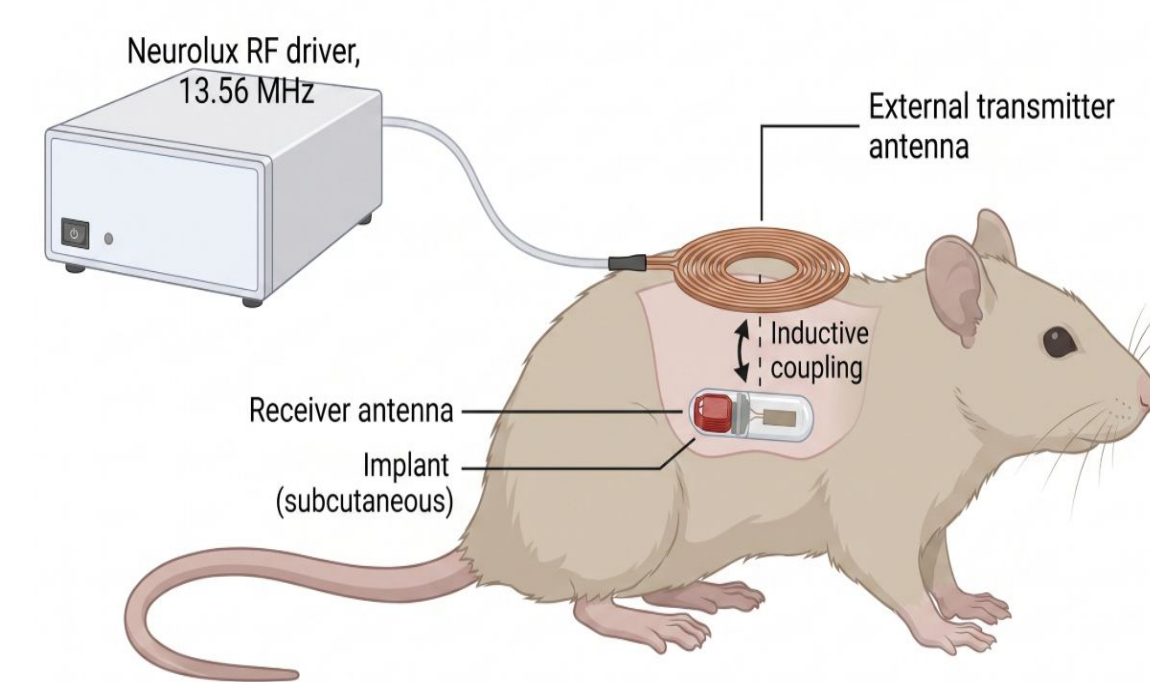


Introduction

- Noncommunicable diseases account for 74% of deaths worldwide, creating a major need for long-term therapeutic strategies.¹ Biologic drugs can effectively treat many chronic conditions, but repeated dosing is costly and burdensome. Engineered-cell implants could instead provide continuous, localized production of therapeutic molecules *in vivo*.
- Engineered living cells offer a fundamentally different approach: cells can be programmed to continuously produce therapeutic proteins *in vivo*, effectively turning an implant into a living drug factory. However, long-term cell function requires encapsulation, adequate oxygen delivery, and supporting hardware - often increasing implant size and invasiveness.²
- Our goal is therefore to enable long-term living therapeutics with a smaller, less invasive implant by providing localized, electrically controlled oxygen generation.³

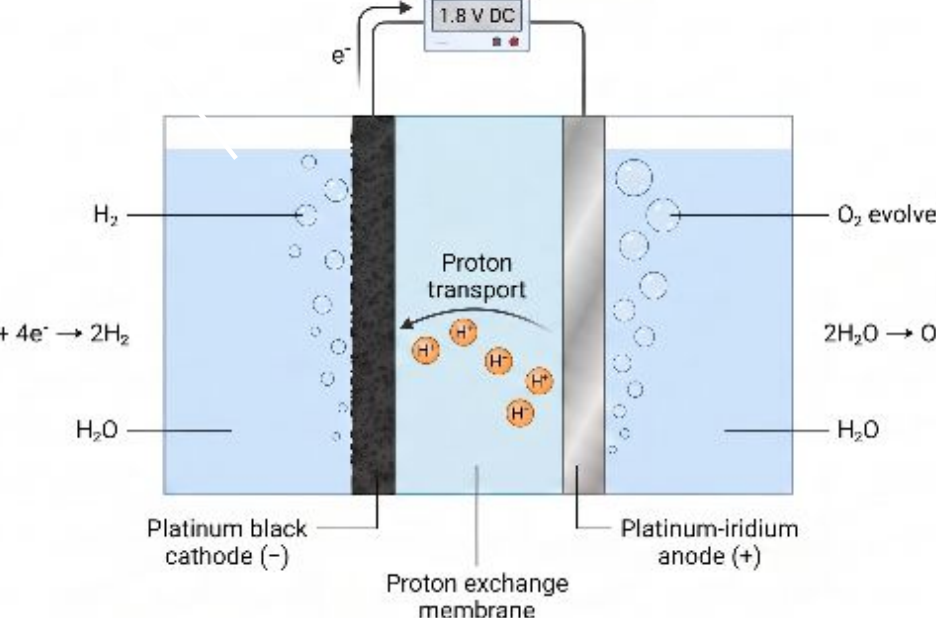
Methods

WPT Design



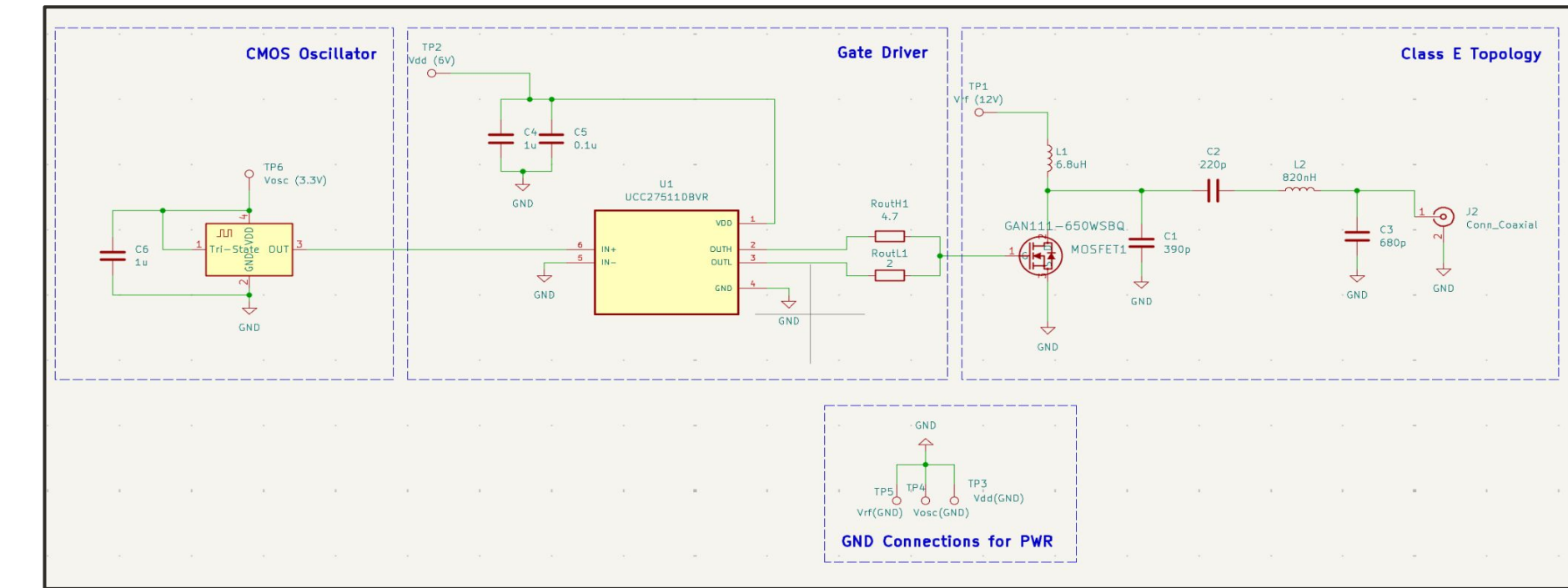
Resonant wireless power transfer uses two coils tuned to the same frequency to efficiently transfer energy through magnetic coupling.

Oxygenator Design



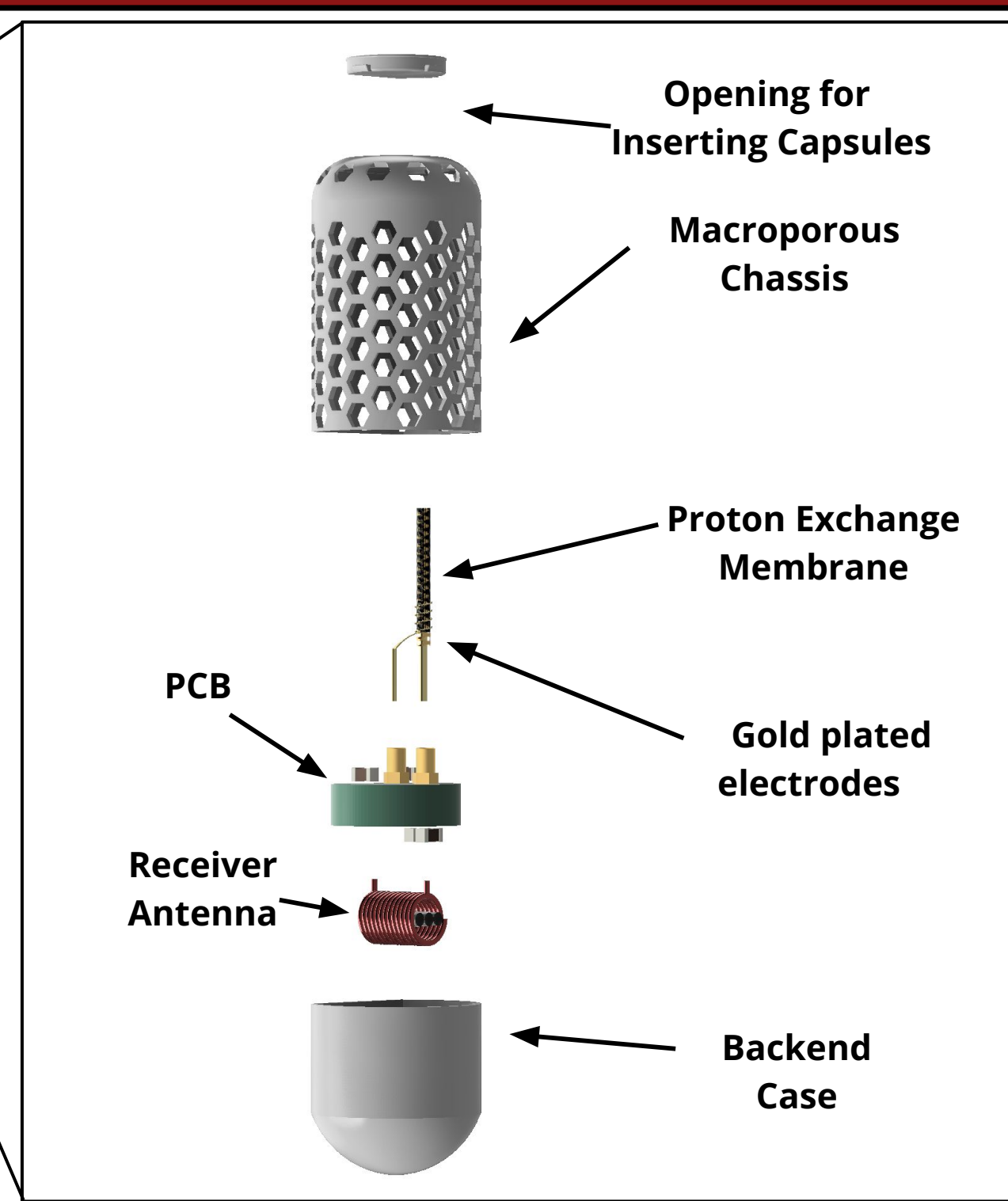
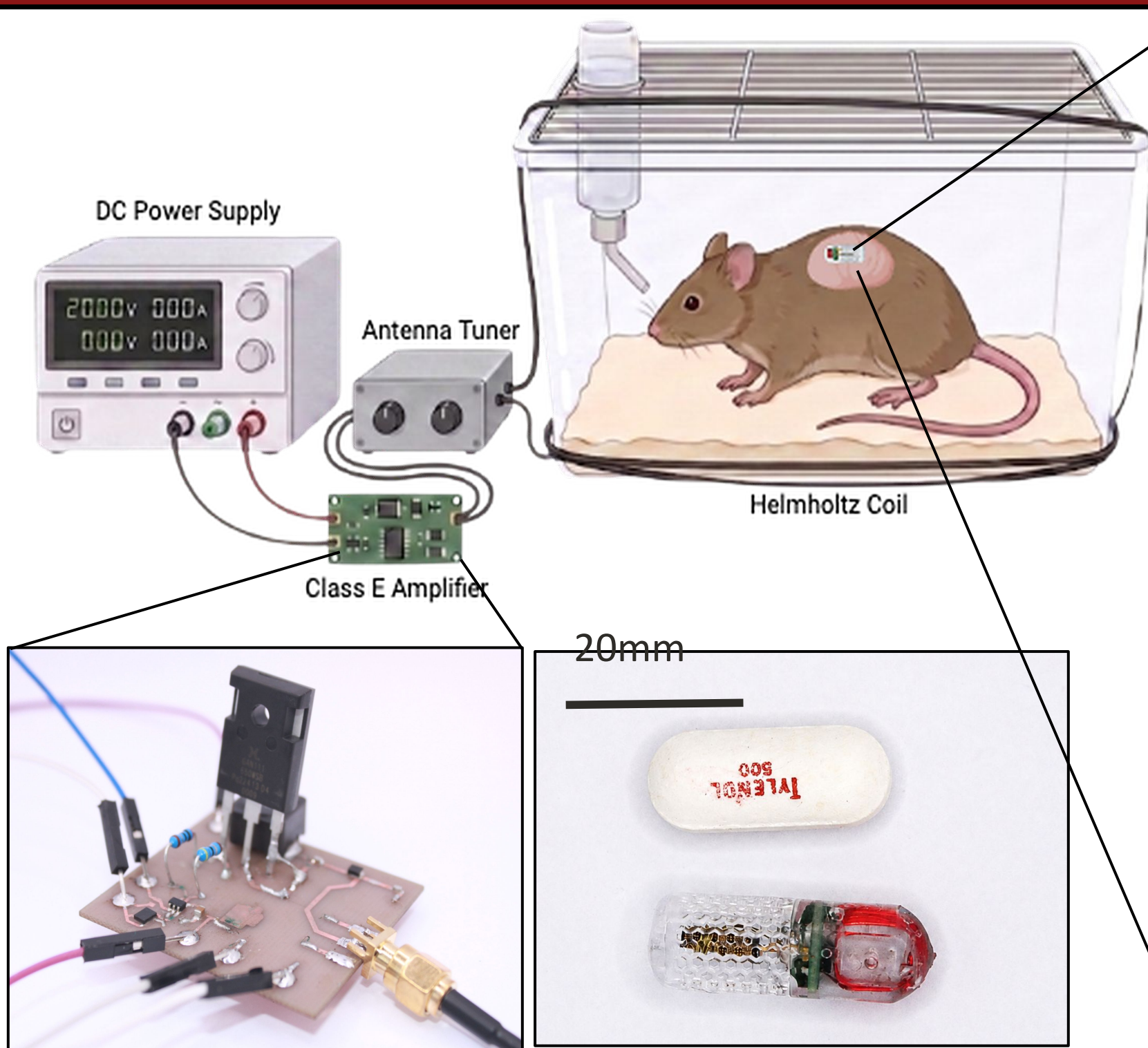
PEM separates the O₂ and H₂ while transporting H⁺ across the membrane, enabling controlled electrochemical oxygen generation.

Class E Power Amplifier Design

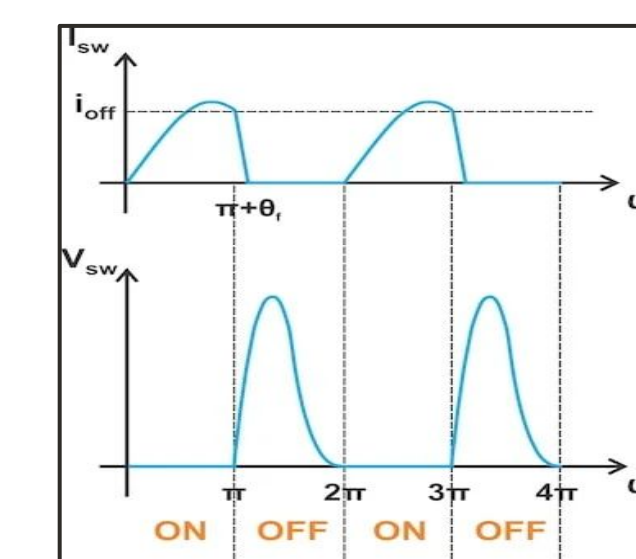


Class E Power Amp is a high-efficiency RF amplifier that takes a low-power RF signal and uses DC supply power to generate a much higher-power RF output at the same frequency.

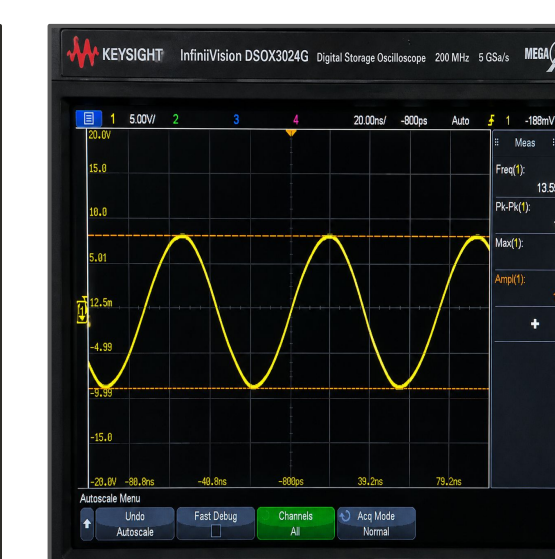
Hardware Design



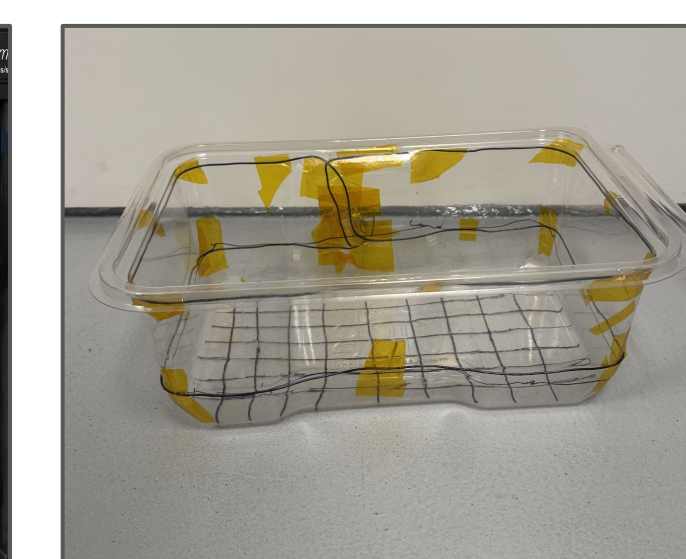
The Class-E amplifier generates a clean 13.56 MHz sinusoidal RF output using low-loss transistor switching, ideally zero voltage switching, and a high-Q resonant network to isolate the signal of interest. The helmholtz transmitter coil converts this RF power into an alternating magnetic field, which induces voltage in the receiver coil by Faraday's law.



ZVS in the transistor switching



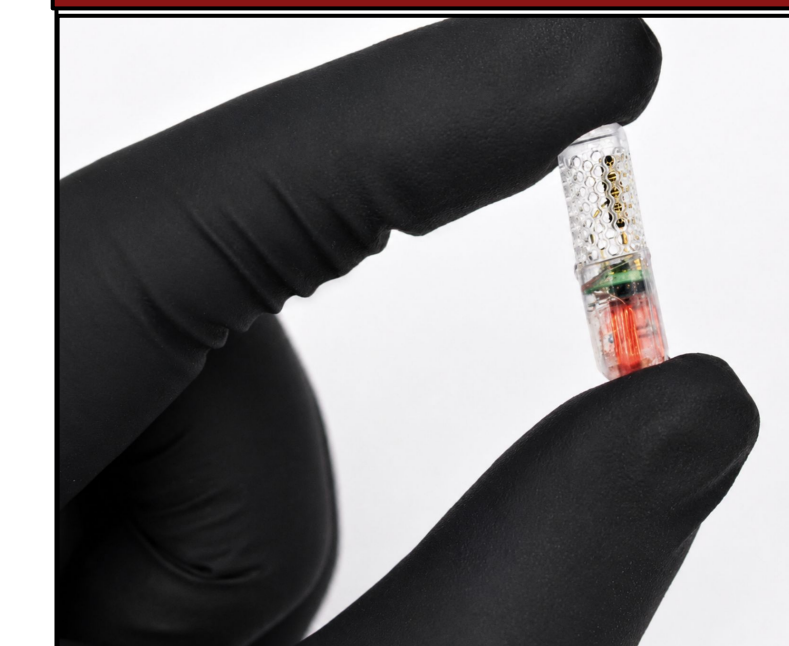
Output Amplified Signal across 50Ω load



Helmholtz Mouse Cage

The device consists of a PCB carrying the tuning network for the receiver antenna, a rectifier, and an LDO regulator, which together convert the received AC signal into a stable 1.8 V output for the oxygenator. Platinum-iridium was selected for the anode as the benchmark acidic OER catalyst, and high-surface-area platinum black for the cathode. The assembly is potted within a housing printed in BioMed Clear resin for biocompatibility. Capsules are loaded through the top opening and secured by a snap-fit lid.

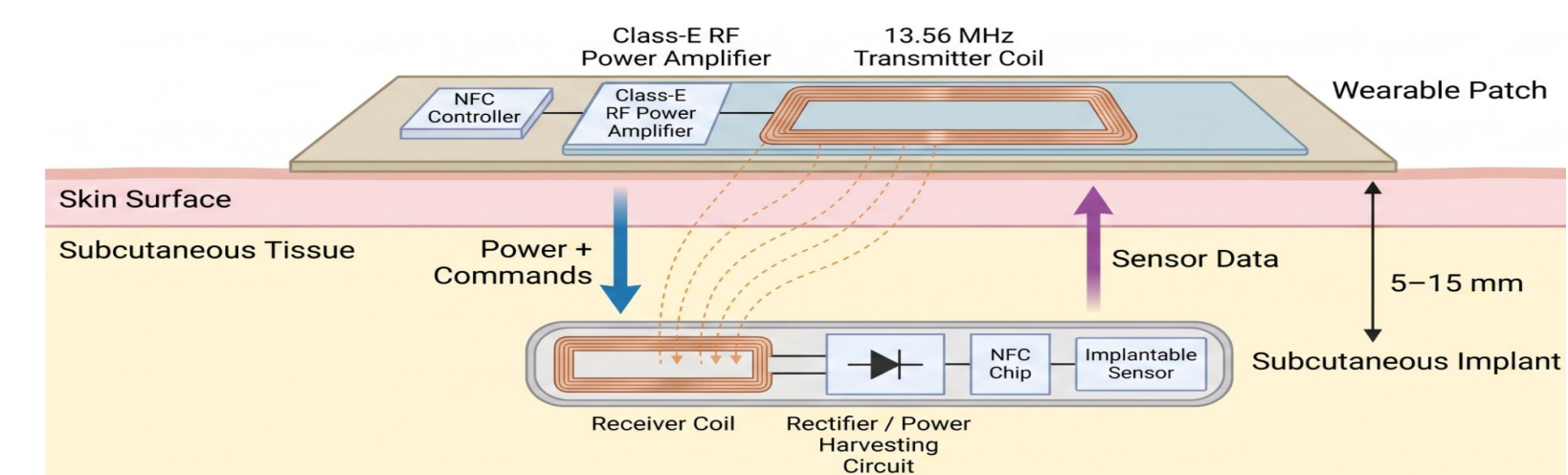
Conclusion



This device provides a wireless way of delivering oxygen to *in-vivo* encapsulated cells that has a chamber that's 5 times as large as previous devices yet 3 times smaller in device size.³

This will allow for *in vivo* studies for:

- Diabetes
- Immunotherapy
- Hormone Replacement Therapy



Wearable, closed-loop wireless power and communication

A wearable transmitter, in the form of a PCB antenna and PCB Class E amp, can provide continuous or intermittent wireless power whilst supporting bidirectional communication, allowing sensor data from the implant to be sent back. This would enable for a closed-loop treatment where implant signals can be used to adjust the delivered power or therapy in real time. The implant could also be powered during bedside charging sessions, reducing the need for an implanted battery.

Acknowledgements

- Thank you to Atharv Naik, our mentor, who helped guide us throughout the entire program.
- Thank you to Dr. Krishnan for his guidance during the summer and Stanford Department of Electrical Engineering for sponsoring our REU.

References

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- [3] S. R. Krishnan et al., "A wireless, battery-free device enables oxygen generation and immune protection of therapeutic xenotransplants *in vivo*," *Proc. Natl. Acad. Sci. U.S.A.*, vol. 120, no. 40, art. e2311707120, 2023, doi: 10.1073/pnas.2311707120.

Power System Test

Figure 1. Comparison of received voltage for the NeuroLux RF driver and custom Class-E amplifier.

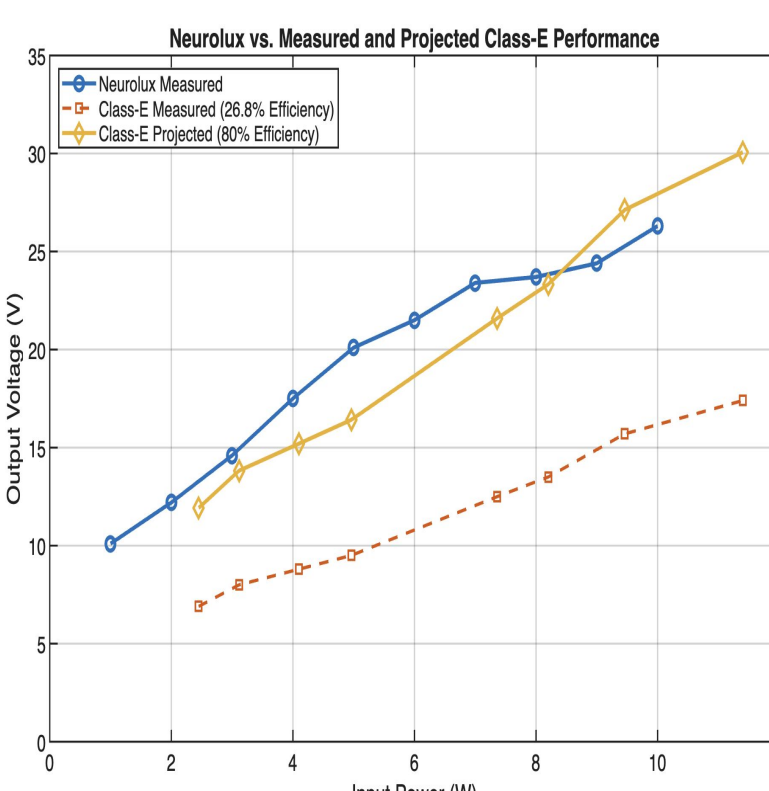


Figure 2. Drain-voltage waveform showing incomplete ZVS, identified as a major source of Class-E switching loss.

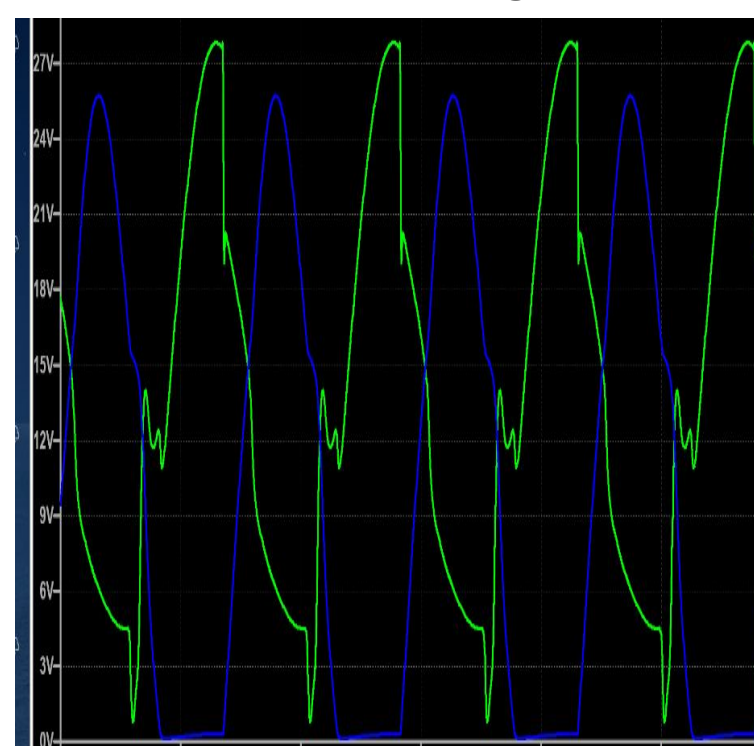
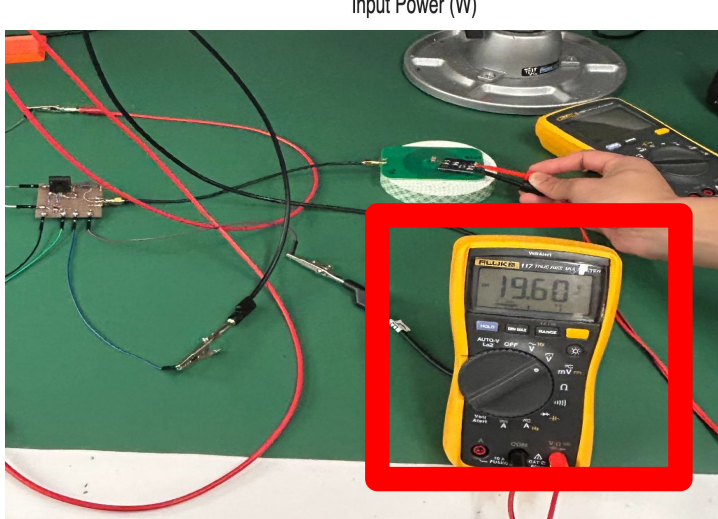
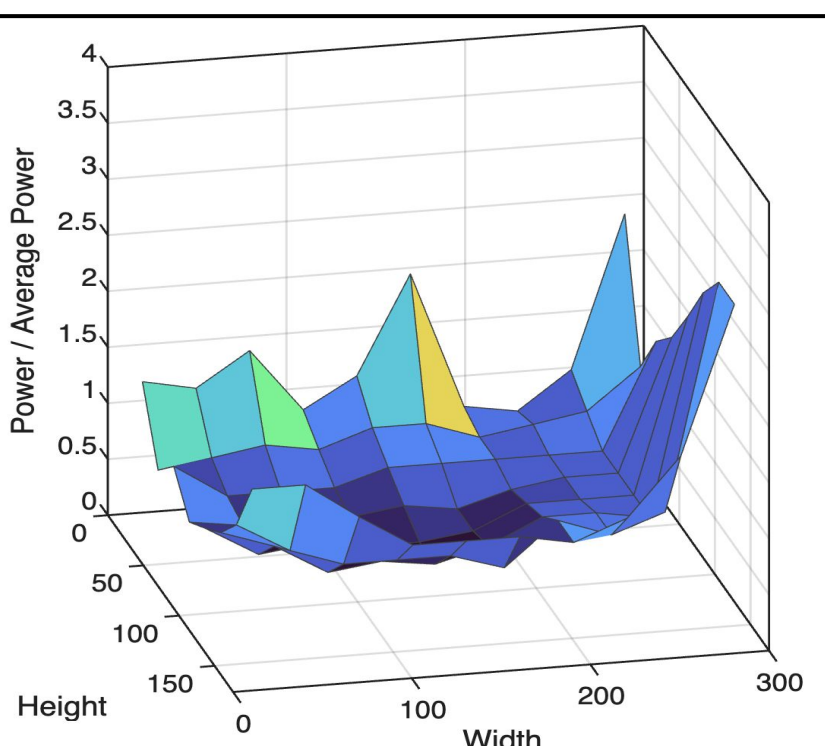


Figure 3. Spatial mapping of received power across the mouse cage at 2 W input.



Benchtop setup to test received power

The NeuroLux RF driver provides reliable wireless power delivery, but costs \$14K. We therefore compared the Class-E system directly against NeuroLux to quantify received power and identify the dominant sources of efficiency loss. Although the Class-E amplifier successfully generated and transferred power at 13.56 MHz, its measured efficiency was below the projected performance. Simulation software showed that the transistor was not consistently achieving ZVS, causing the device to turn on while a significant drain voltage was still present. This voltage-current overlap increases switching loss and is a major contributor to the reduced efficiency.

Oxygen Results

Figure 4. Device-to-device variation in electrolysis current at 1.8 V

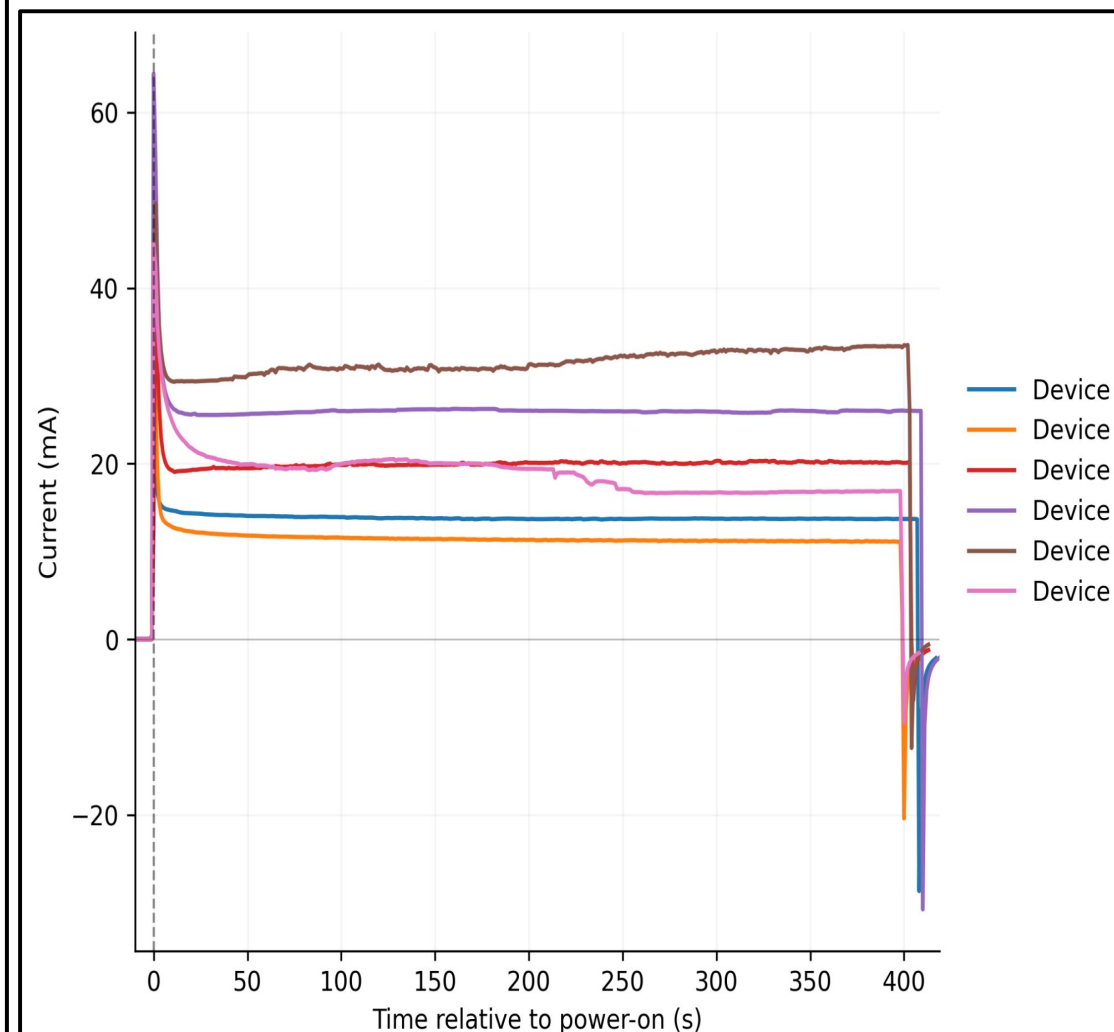
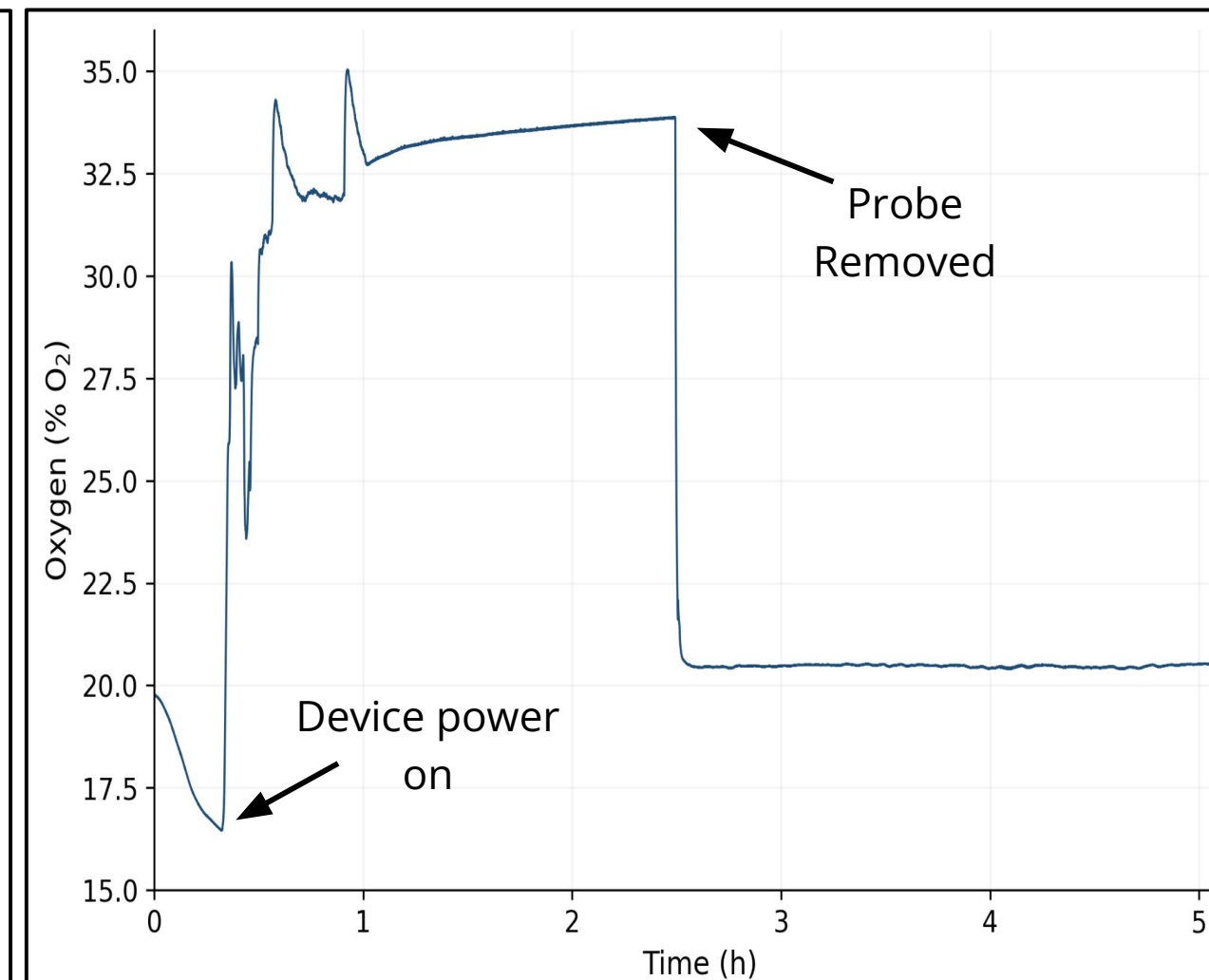


Figure 5. In vitro O₂ generation in a water bath



In Figure 4, six oxygenation devices were connected to a benchtop DC power supply and submerged in DI water. The devices were turned on for 400s and the resulting current was measured. In Figure 5, a fully assembled device was placed in DI water and powered wirelessly through resonant inductive coupling. An oxygen probe was inserted into the capsule chamber and allowed to equilibrate. The device was then operated for 2 h before the probe was withdrawn from the capsule and left in the surrounding water.